

A Adjusted EF definition

$$EF = \frac{|G \cap H|}{(N_L + N_{\bar{U}}) h_{rand}}$$

$$h_{rand} = |H|/|L|$$

- U all valid genes
- L genes measured in this screen
- H hits in the screen library
- G genes acquired over N rounds

B Effective-budget terms

$$N_L = |G \cap L|$$

acquired genes measured in the screen

$$N_{\bar{U}} = |G \setminus U|$$

invalid or hallucinated acquisitions

$$n_{eff} = N_L + N_{\bar{U}}$$

total effective budget used

C Worked comparison: raw hits versus enrichment

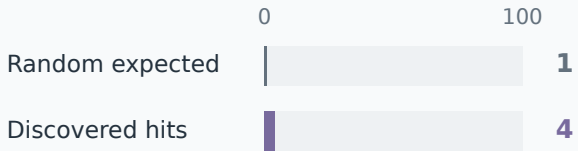
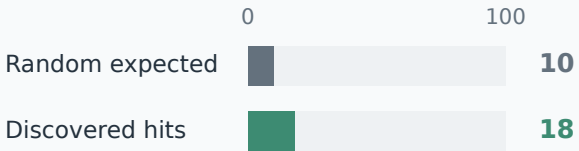
Screen A

- **Hit-Rich:** $h_{rand} = 10\%$
- **In a batch of 100 genes,** random expects **10 hits.**

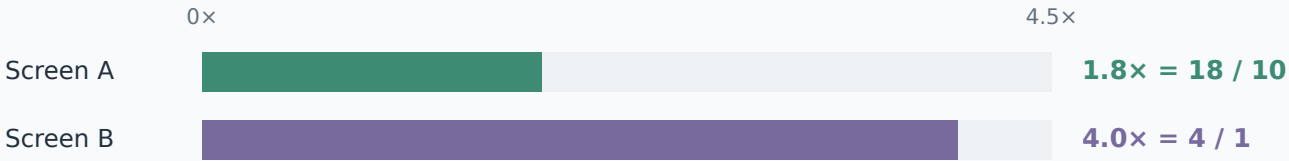
Screen B

- **Hit-Sparse:** $h_{rand} = 1\%$
- **In a batch of 100 genes,** random expects **1 hit.**

Raw hits in a 100-gene batch

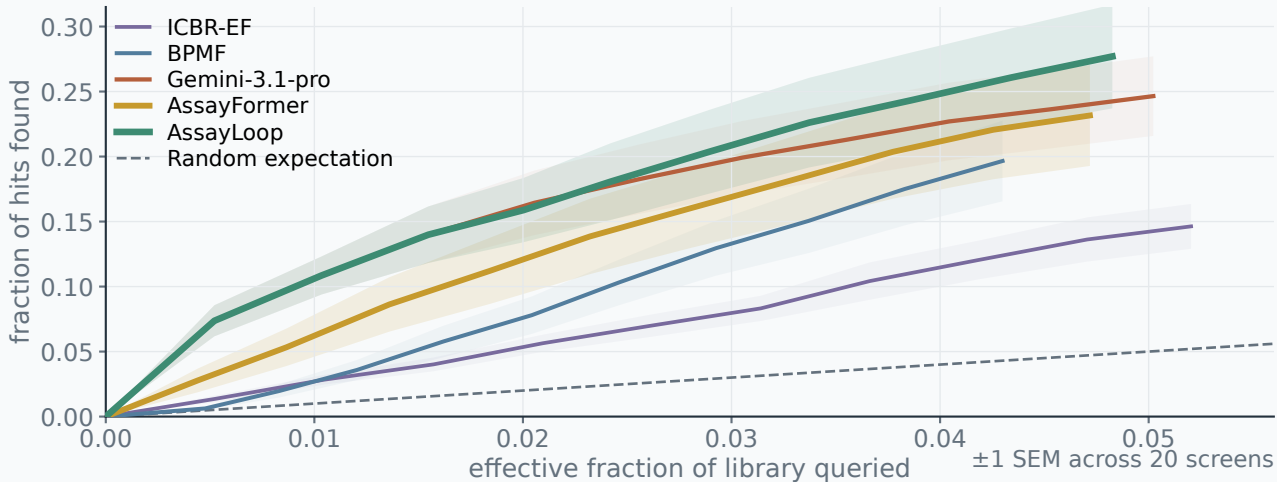


Enrichment factor

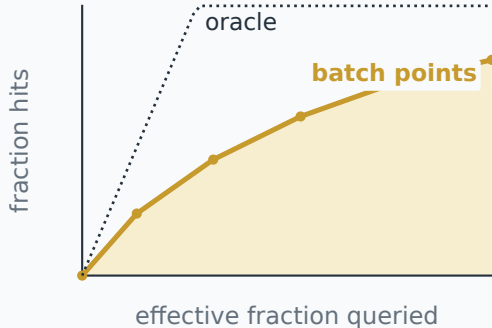


More raw hits can still represent less enrichment over random.

D Actual recovery trajectories



E Adjusted nAUC definition



$$x_t = \frac{\sum_{i=0}^t (N_{L,i} + N_{\bar{U},i})}{|L|} \quad y_t = \frac{\sum_{i=0}^t N_{H,i}}{|H|}$$
$$nAUC = \frac{AUC}{AUC_{oracle}}$$

F Complementary diagnostics

- Fraction of hits**
|G ∩ H|/|H|
endpoint yield
- Shortfall**
|G ∖ L|/|G|
unscoreable proposals
- % essential**
|G ∩ H ∩ E|/|G ∩ H|
universal vs specific hits
- Vendi ratio**
effective diversity / batch
higher = more distinct
- Pathway overlap**
pairwise Jaccard / random
>1 = more redundant
- Effective pathways**
EP = exp(H(p))
higher = broader coverage